



# Ruby AI Tutor

System Architecture & Technical Design Document

• Live on Vercel · Production

2

AI ENGINES

106

ADAPTIVE SKILLS

11

API ROUTES

34

READING SKILLS

72

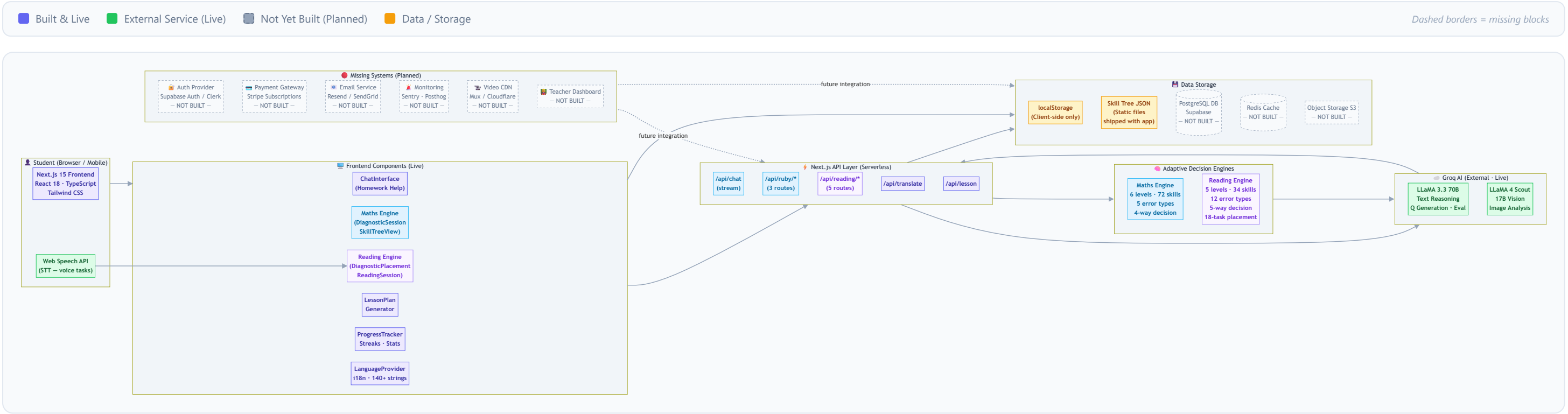
MATHS SKILLS

18

DIAGNOSTIC TASKS

Next.js 15 · React 18 · TypeScript · Tailwind CSS  
Groq LLaMA 3.3 70B · Vercel Edge · Web Speech API

Prepared for Lead Developer Review · March 2026



Built & Live Not Yet Built

LAYER 1 — CLIENT / FRONTEND · NEXT.JS 15 · REACT 18 · TYPESCRIPT · TAILWIND CSS · DEPLOYED ON VERCEL

HomeScreenChatInterface (image upload)LessonPlanProgressTracker + StreaksSettingsViewOnboardingFlowSidebar / TopBarLanguageProvider (i18n)SpinningGlobe (translation UI)

MATHS ENGINE COMPONENTS

DiagnosticSessionQuestionCardFeedbackCardSkillTreeView (6 levels · 11 tiers · 72 skills)StudentDashboard

READING ENGINE COMPONENTS

ReadingSessionReadingDiagnosticPlacement (D01–D18)ReadingSkillTreeView (5 levels · 14 tiers · 34 skills)

NOT YET BUILT

Video Learning ("Watch" mode)Teacher DashboardParent PortalPDF Report DownloadServiceWorker / PWA Offline

HTTPS / REST / Server-Sent Events (streaming)

LAYER 2 — BACKEND API · NEXT.JS APP ROUTER SERVERLESS ROUTES · VERCEL EDGE FUNCTIONS

POST /api/chatPOST /api/lessonPOST /api/translatePOST /api/ruby/generate-questionPOST /api/ruby/submit-answerPOST /api/ruby/student-profilePOST /api/reading/diagnosticPOST /api/reading/diagnostic/evaluatePOST /api/reading/diagnostic/placementPOST /api/reading/generate-questionPOST /api/reading/submit-answer

NOT YET BUILT

/api/auth/\*/api/user/\*/api/billing/\*/api/reports/\*/api/teacher/\*/api/webhooks/stripe

SDK calls / JSON

LAYER 3 — ADAPTIVE DIAGNOSTIC ENGINES (CLIENT-SIDE LOGIC · LIB/)

Maths Engine (lib/student-model.ts · data/skill-tree.json)

- › 6 levels · 11+ tiers · 72 atomic skills
- › 3 question templates: concrete / story / symbolic
- › 5 error types: conceptual gap · strategy gap · representation confusion · execution slip · correct
- › Mastery gate: 3 correct in 2+ question formats
- › 4-way decision: ADVANCE · PRACTICE · RETEACH · BACKTRACK

Reading Engine (lib/reading-student-model.ts · data/reading-skill-tree.json)

- › 5 levels · 14 tiers · 34 atomic skills
- › 18-task diagnostic placement (D01–D18)
- › Hard Gate 1: CVC encoding <0.8 → Level 3 block
- › 12 error types: ERR\_PHONEME\_CONF · ERR\_BLEND\_FAIL · ERR\_VOWEL\_CONF · ERR\_SIGHT\_MISS...
- › 5-way decision: ADVANCE · PRACTICE · RETEACH · BACKTRACK · ACCELERATE

Groq SDK · HTTPS

LAYER 4 — EXTERNAL SERVICES

Groq API — LLaMA 3.3 70B (text)Groq API — LLaMA 4 Scout 17B (vision)Vercel (hosting · CDN · CI/CD)GitHub (source control · auto-deploy)Web Speech API (browser STT)

NOT YET BUILT

Supabase Auth / Clerk (authentication)Stripe (payments)Resend / SendGrid (email)Sentry (error monitoring)Posthog (analytics)Mux / Cloudflare Stream (video)

Read / Write

LAYER 5 — DATA PERSISTENCE

localStorage — student profiles · chat history · mastery · streaks · language · dark modeStatic JSON — skill-tree.json · reading-skill-tree.json

NOT YET BUILT

PostgreSQL (Supabase) — persistent user data across devices

Redis — session cache · rate limiting

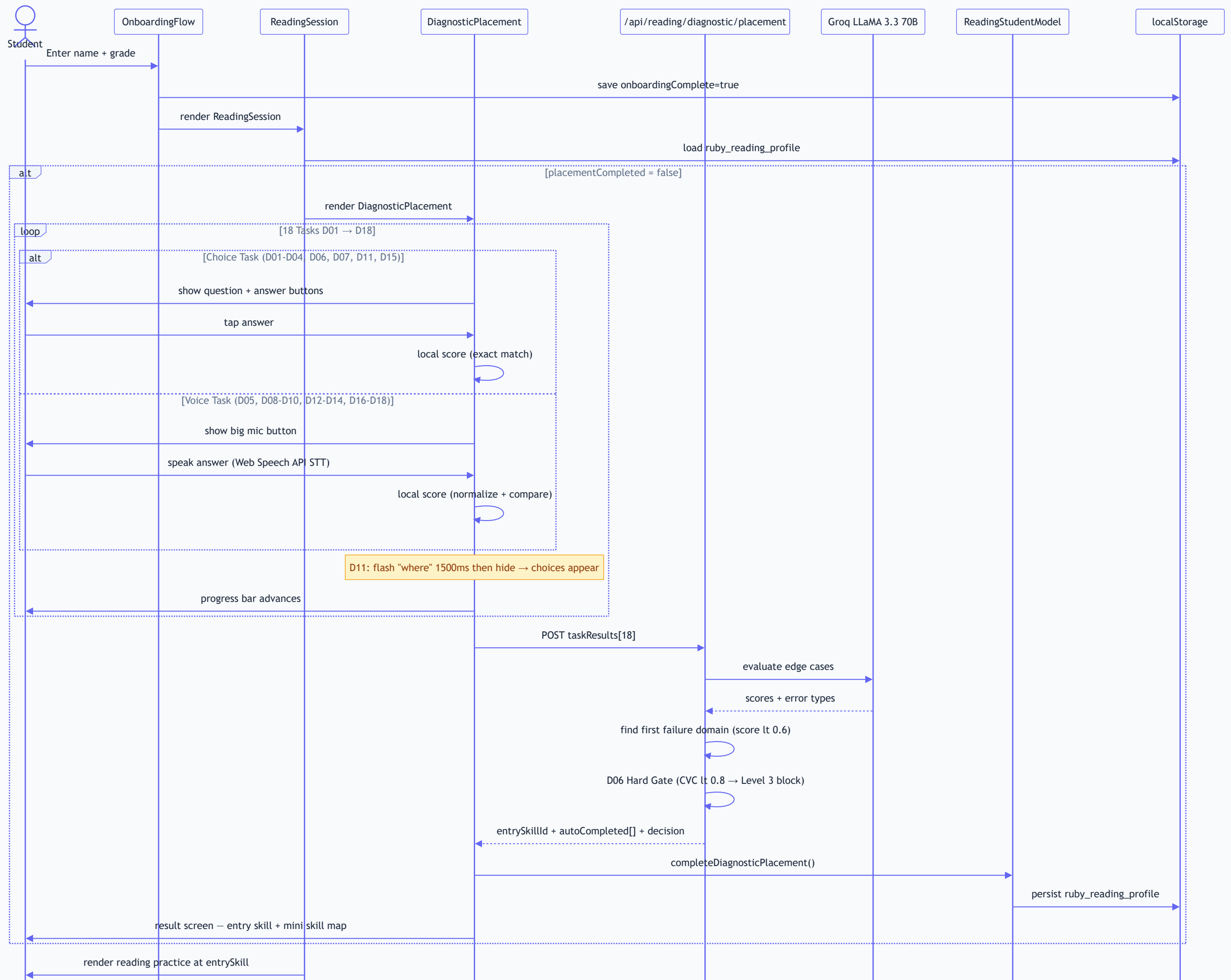
S3 / Object Storage — images · PDFs · audio

PostgreSQL (Supabase) — persistent user data across devices

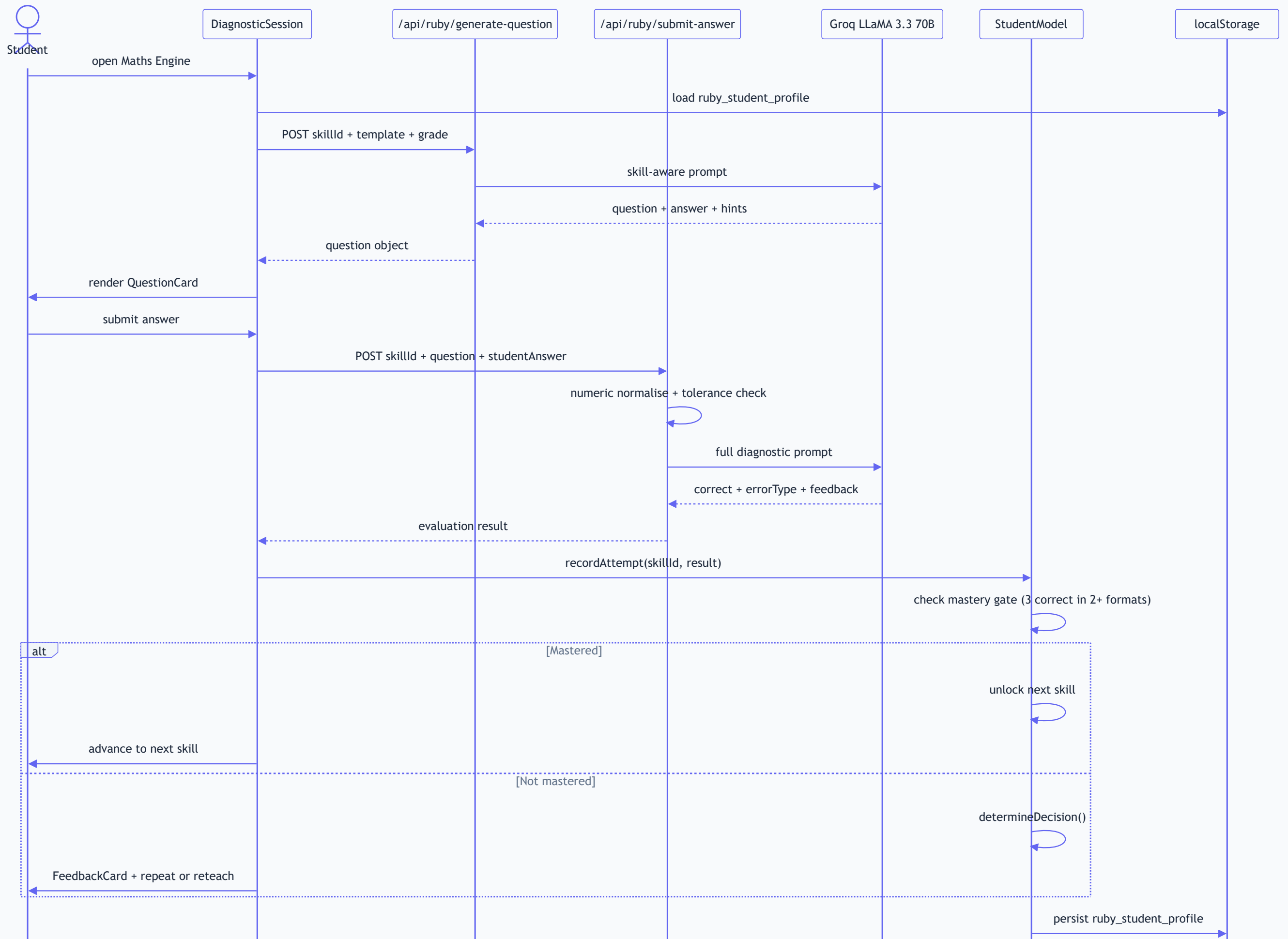
Redis — session cache · rate limiting

S3 / Object Storage — images · PDFs · audio

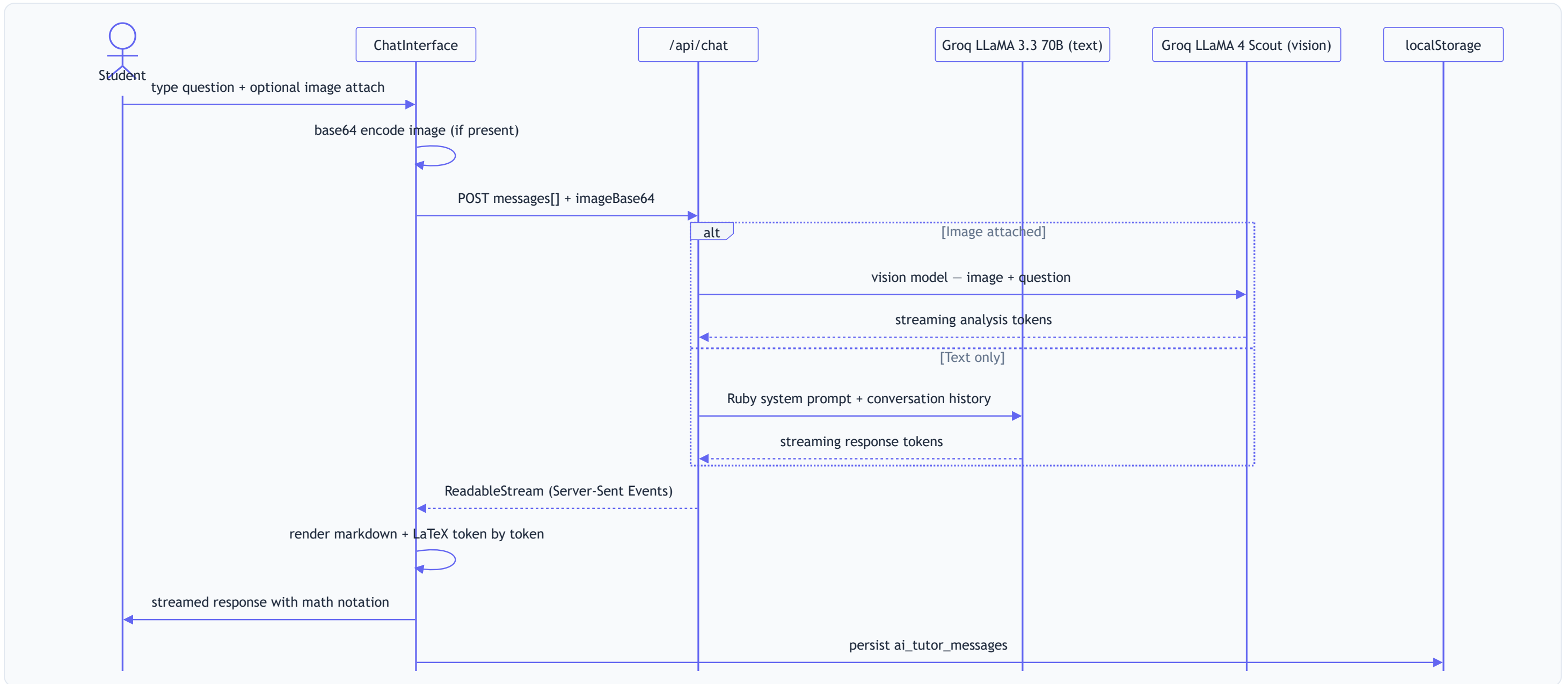
**1 Student Onboarding → Reading Diagnostic Placement (D01–D18)**





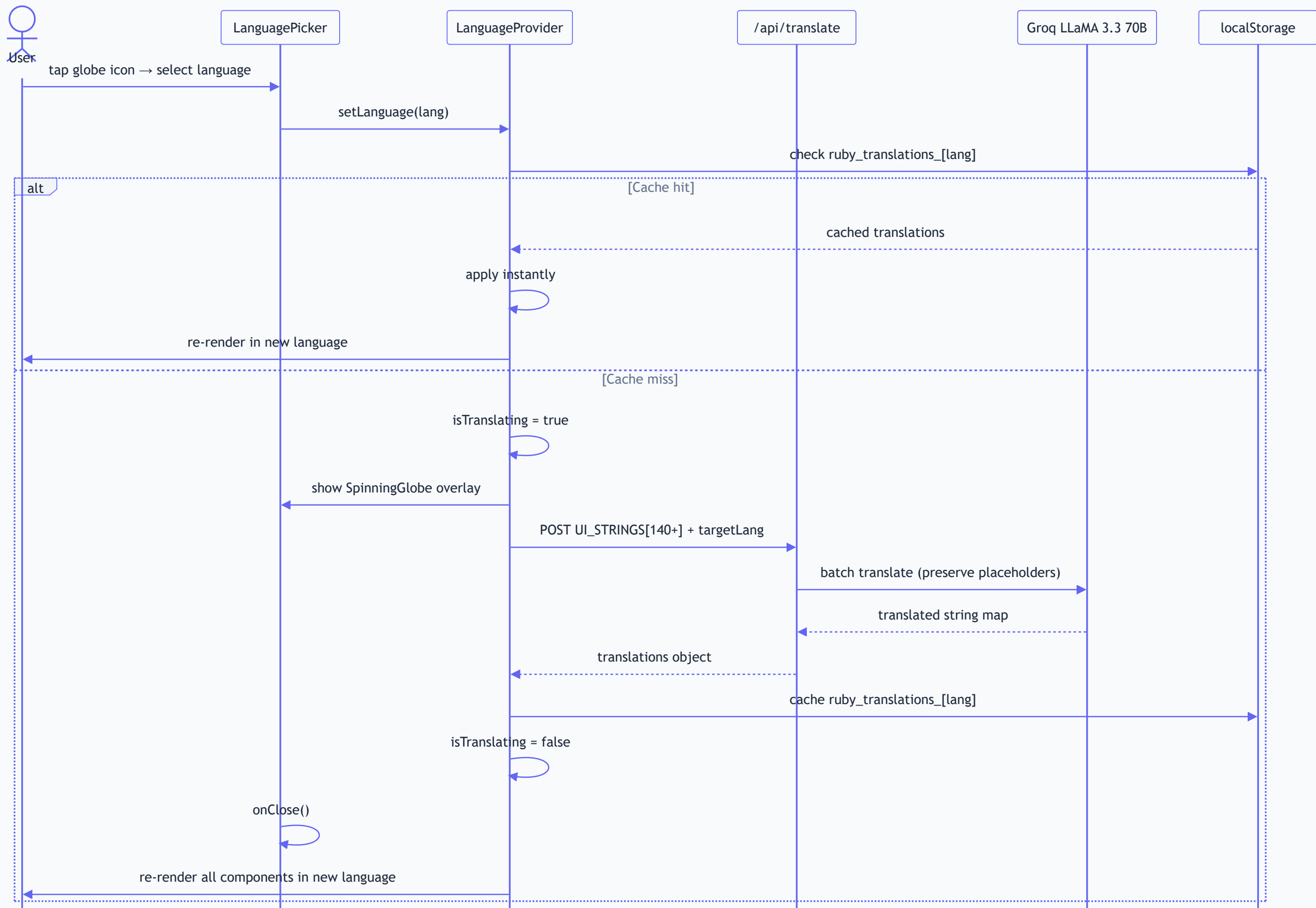


3 Homework Chat with Image Upload (Streaming)



4 Platform Language Switch — i18n Translation Flow





The blocks below are designed in the UI (settings billing page, video placeholder, download report button) but have no backend implementation yet. All grey dashed = not built.

Authentication System

No login, no user accounts. Students identified only by localStorage. Loses progress on device switch. Recommended: Supabase Auth or Clerk for email + Google OAuth.

Backend Database

All profile, mastery, and session data stored in browser localStorage. Resets on clear. Target: Supabase PostgreSQL with tables for users, skill\_mastery, sessions, error\_history.

Payment Gateway

Billing UI exists in Settings but no integration. No subscription plans enforced. Target: Stripe Checkout + Customer Portal + webhook handlers for plan gating.

Video Learning

"Watch" mode in sidebar shows "Coming soon" placeholder. Target: Mux or Cloudflare Stream for low-latency adaptive video. Curriculum-aligned videos per skill tier.

Analytics Platform

No behavioural tracking beyond local streak data. No funnel analysis, no drop-off detection. Target: Posthog (open-source) for user behaviour + learning analytics dashboard.

Error Monitoring

No production crash reporting. API errors silently swallowed. Target: Sentry for real-time error tracking, performance monitoring, and API failure alerts.

Email Service

No transactional email. No welcome email, weekly progress summary, or subscription receipts. Target: Resend (modern API) for automated parent/student emails.

PDF Progress Reports

Download button in Settings does nothing. Target: React PDF or Puppeteer-based report generation covering skill mastery, error patterns, and weekly progress.

Teacher Dashboard

Single-student app only. No class management, no multi-student view, no teacher login. Required for school licensing. Target: separate teacher role + class overview UI.

Architecture Gap: Current vs Target

CURRENT STATE

MVP — Single Device, No Auth

Browser → Next.js frontend → Serverless API routes → Groq LLM

Data: localStorage (browser only)

Auth: none

Billing: none

Scale: single user per browser

TARGET STATE

Production SaaS — Multi-tenant Platform

Browser/App → Next.js → API → Groq LLM + Supabase DB

Auth: Supabase Auth (student · parent · teacher roles)

Billing: Stripe (Free · Pro · School plans)

Scale: thousands of concurrent students

Monitoring: Sentry · Posthog · Resend

PHASE 1 · FOUNDATION

Auth + Database

1

Set up Supabase project + PostgreSQL schema (users, sessions, skill\_mastery, error\_history)

2

Migrate localStorage reads/writes to Supabase client in lib/student-model.ts + lib/reading-student-model.ts

3

Add Supabase Auth — email/password + Google OAuth, student/parent roles

4

Protect all API routes with JWT middleware (supabase.auth.getUser)

5

Add Sentry SDK — capture API errors + client crashes in production

6

Deploy updated schema to Vercel with Supabase env vars

PHASE 2 · MONETISATION

Billing + Reports

1

Stripe integration — Free (3 skills/day) · Pro (unlimited) · School (class management) plans

2

Stripe Customer Portal + /api/webhooks/stripe for subscription lifecycle

3

Wire Settings billing page to real Stripe checkout session

4

PDF progress report generation (React PDF library) — weekly summary, skill map, error analysis

5

Resend email — welcome, weekly digest, subscription receipts

6

Posthog analytics — funnel tracking, skill drop-off, session length

PHASE 3 · SCHOOL PLATFORM

Scale + Video

1

Teacher Dashboard — class management, per-student skill view, assignment setting

2

Parent portal — weekly progress summaries, streak notifications

3

Video Learning module — Mux streaming, curriculum-aligned per skill tier

4

Offline PWA — service worker + IndexedDB for offline skill practice + sync on reconnect

5

S3 object storage — homework photo uploads, audio session recordings, PDF archive

6

Redis cache — session tokens, rate limiting, frequent student profile reads

Recommended Tech Stack — Full Target Architecture

| LAYER          | CURRENT (MVP)                    | TARGET (PRODUCTION)             | STATUS    |
|----------------|----------------------------------|---------------------------------|-----------|
| Frontend       | Next.js 15 + React 18 + Tailwind | Same (add PWA service worker)   | ✓ Live    |
| AI Model       | Groq LLaMA 3.3 70B + Vision      | Same (add fine-tuning pipeline) | ✓ Live    |
| Database       | localStorage (browser only)      | Supabase PostgreSQL             | ⚠ Phase 1 |
| Authentication | None                             | Supabase Auth / Clerk           | ⚠ Phase 1 |
| Payments       | None (UI stub only)              | Stripe Subscriptions            | ✗ Phase 2 |
| Video          | Placeholder "coming soon"        | Mux / Cloudflare Stream         | ✗ Phase 3 |
| Monitoring     | None                             | Sentry + Posthog                | ⚠ Phase 1 |
| Email          | None                             | Resend                          | ✗ Phase 2 |

